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ORIGINAL ARTICLE

Predictors of 1-year outcome after cardiac re-transplantation: Machine learning analysis

Atsushi Kainuma[1] | Yuming Ning[2] | Paul A. Kurlansky[1,2] | Amy S. Wang[1] | Farhana Latif[3] | Gabriel T. Sayer[3] | Nir Uriel[3] | Yuji Kaku[1] | Yoshifumi Naka[1] | Koji Takeda[1]

[1]Division of Cardiothoracic Surgery, Department of Surgery, Columbia University Medical Center, New York, New York, USA

[2]Center for Innovation and Outcomes Research, Columbia University, New York, New York, USA

[3]Department of Medicine/Cardiology, Columbia University Medical Center, New York, New York, USA

Correspondence
Koji Takeda, 177 Fort Washington Avenue, New York, NY 10032, USA.
Email: kt2485@cumc.columbia.edu

Abstract

Background: As cardiac re-transplantation is associated with inferior outcomes compared with primary transplantation, allocating scarce resources to appropriate re-transplant candidates is important. The aim of this study is to elucidate the factors associated with 1-year mortality in cardiac re-transplantation using the random forests algorithm for survival analysis.

Methods: We retrospectively reviewed the United Network for Organ Sharing registry and identified all adult (> 17 years old) recipients who underwent cardiac re-transplantation between January 2000 and March 2020. The random forest algorithm on Cox modeling was used to calculate the variable importance (VIMP) of independent variables for contributing to 1-year mortality.

Results: A total of 1294 patients underwent cardiac re-transplantation. Of these, 137 patients were re-transplanted within 1 year of their first transplant, while 1157 patients were re-transplanted more than 1 year after their first transplant. One-year mortality was significantly higher for patients receiving early transplantation compared with those receiving late transplantation (Early 40.6% vs. Late 13.6%, log-rank $P < .001$). Machine learning analysis showed that total bilirubin (> 2 mg/dl) (VIMP, 2.99%) was an independent predictor of 1-year mortality after early re-transplant. High BMI (> 30.0 kg/m^2) (VIMP, 1.43%) and ventilator dependence (VIMP, 1.47%) were independent predictors of 1-year mortality for the late re-transplantation group.

Conclusion: Machine learning showed that optimal 1-year survival following cardiac re-transplantation was significantly related to liver function in early re-transplantation, and to obesity and preoperative ventilator dependence in late re-transplantation.

KEYWORDS
heart (allograft) function/dysfunction, heart transplantation, machine learning, retransplantation, United Network for Organ Sharing (UNOS)

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1 | INTRODUCTION

Although heart transplant outcomes have improved dramatically, patients with end-stage heart failure who receive heart transplants still suffer from early and late graft failure. Several therapeutic options have been proposed to overcome graft failure, but cardiac retransplantation is still considered the gold standard in terms of long-term outcomes.[1]

The number of patients who are candidates for cardiac retransplantation has increased.[2–4] However, the number of cardiac retransplants performed has remained steady over time, between 2 and 4% of all adult heart transplants.[5,6] The global shortage of available donor organs and the poor clinical outcomes of re-transplant patients raise ethical and economic concerns. As cardiac re-transplantation is associated with inferior outcomes compared with primary transplantation, allocating scarce resources to the most appropriate retransplant candidates may help optimize the use of a scarce and vital resource.

In a recent study, the outcomes of patients who underwent late re-transplantation (> 1 year after previous transplant) seem to be comparable to the outcomes of patients who underwent primary cardiac transplantation.[7,8] In contrast, early re-transplantation (≦1 year after previous transplant) is associated with lower 1-year survival compared with late re-transplantation.[6–8] Appropriate assessment of indications and comorbidities is essential when evaluating candidates for re-transplantation. However, information on variables related to survival after re-transplantation is still limited.

Machine learning models have been effective at predicting risk factors for various cardiovascular diseases.[9–12] To better understand the relationship between variables related to post-transplant survival, we used machine learning analysis via random survival forests (RSF). One major advantage of RSF is the accuracy of prediction. Breiman et al. has shown that the prediction of ensemble learning can be substantially improved by random forests.[13] Ishwaran et al. implemented and extended Breiman's random forests method into RSF to investigate important factors related to breast cancer using RSF.[14] However, no coefficient estimates is a main disadvantage of RSF that should be noted. Therefore, we performed a complementary Cox analysis to verify the results. The aim of this study is to elucidate risk factors associated with 1-year mortality in cardiac re-transplantation.

2 | MATERIAL AND METHODS

Approval with waiver of consent was granted by the Institutional Review Board of Columbia University Medical Center in New York, New York. We retrospectively reviewed the United Network for Organ Sharing (UNOS) registry and identified all adult (> 17 years old) cardiac re-transplant recipients between January 2000 and March 2020. Recipients were divided into early (≦1 year after previous transplant) and late (> 1 year after previous transplant) groups based on the timing of re-transplantation (Figure 1). A period of 1 year was chosen based on previous definitions.[7,15,16] Univariate spline regression analyses were performed in order to further examine the relationship between time from previous transplant to retransplantation and 1-year mortality. To explore temporal trends in retransplantation patient selection, patient outcome, and risk markers, patients were divided based on whether they were transplanted before or after the UNOS policy changes (January 1, 2000–October 17, 2018 or October 18, 2018–March 18, 2020).[17,18] The time before the rule change was divided into two periods: January 2000 to December 2009 (the early pre-policy era), and January 2010 to October 2018 (the late pre-policy era).

The following information at the time of transplant was collected regarding recipient: age, gender, body mass index (BMI), ethnicity, number of previous transplantation, history of malignancy, smoking history, diabetes, creatinine, dialysis after listing, total bilirubin (tBil) level, need for transfusion after listing, extracorporeal membrane oxygenation (ECMO), intra-aortic balloon pump (IABP), left ventricular assist device (LVAD), total artificial heart (TAH), ventilator dependence, inotropic therapy dependence, pulmonary capillary wedge pressure (PCWP), pulmonary vascular resistance (PVR), panel reactive antibody (PRA), functional status, and intensive care unit (ICU) stay. The following information regarding donors was collected: age, gender, BMI, donor risk index,[19] blood urea nitrogen/creatinine (BUN/Cre), donor ethnicity, history of hepatitis C, the Center for Disease Control high-risk (CDC HR) donor, cocaine or other intravenous drug usage, donor ejection fraction, donor hypertension, donor diabetes, smoking and ischemic time.[20–24] The donor factor with a donor risk score above the 90th percentile of risk, which corresponds to a score of 7 or higher, was also considered.[25]

3 | STATISTICAL ANALYSIS

All continuous variables were checked for normality of distribution by using the Shapiro-Wilk test and found not to be normally distributed and therefore are represented as median (interquartile range, IQR) and compared using the Wilcoxon rank-sum test. Categoric variables were summarized by frequency and percentage and were compared between the groups by the $\chi 2$ test or the Fisher exact test.

The variables selected for inclusion in the model were those known to be associated with risk of mortality after OHT.[7,20–22,26,27] Missing data can significantly impact the results of variable importance, so variables with more than 10% missing data (functional status, PCWP, CDC HR donor, recipient smoking history, PVR, and PRA) were excluded from the machine learning analysis (Tables S1 and S2).

An R package "RandomForestSRC" was also applied to identify important predictors of 1-year mortality using the variables mentioned previously.[14] The random forest algorithm on Cox modeling was used to calculate the variable importance of the independent variables for the time to 1-year mortality. We also ran multivariable Cox modeling on 1-year mortality by using the risk factors mentioned previously (backward selection; $P < .20$ to stay). Baseline demographics variables shown to be significantly associated with postoperative mortality ($P < .05$) in univariate analysis or of clinical interest were entered into multivariate analysis. Survival was estimated using the Kaplan-Meier method and was compared across groups by the log-rank test. All analyses were performed using SAS software version 9.4 (SAS Institute Inc) and

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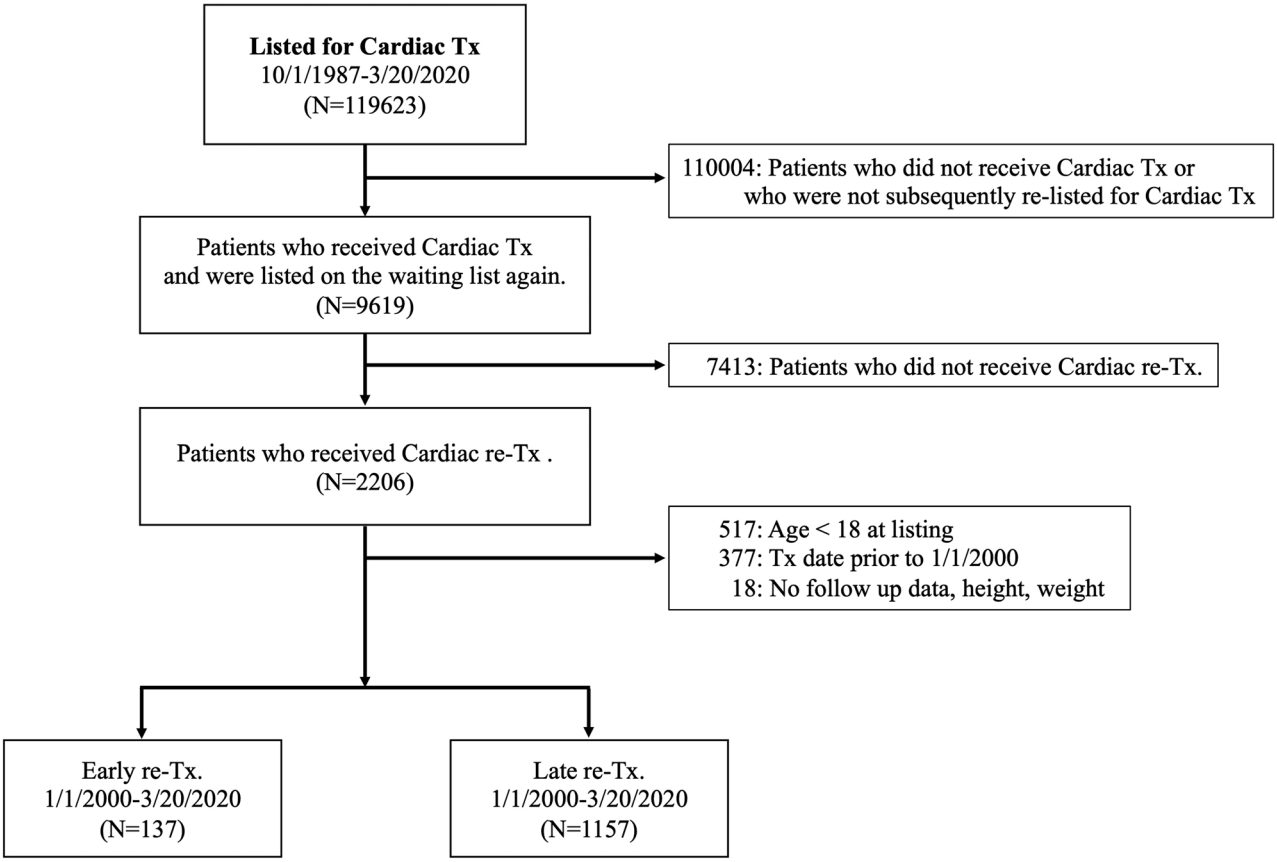

FIGURE 1 Study flow chart. Tx, transplantation

R software version 4.0.3 (R Foundation). Two-sided *P* values less than .05 were considered statistically significant.

4 | RESULTS

4.1 | Baseline characteristics

There were 137 recipients in the early cardiac re-transplantation group and 1157 recipients in the late cardiac re-transplantation group. Over the study period, the number of late cardiac re-transplants increased slightly, while the number of early cardiac re-transplants remained stable (Figure S1).

Baseline characteristics are shown in Tables 1 and 2. Recipients in the early cardiac re-transplantation group were more likely to need dialysis (28.9% vs. 13.1%, $P < .001$) and blood transfusion after listing (38.9% vs. 17.6%, $P < .001$).

At transplant, compared to the recipients in the late cardiac retransplantation group, the recipients receiving early cardiac re-transplantation were older (50 years vs. 44 years, $P = .004$), had higher tBil levels (36.6% vs. 9.1%, $P < .001$) and PVR (2.1 WU vs. 1.5 WU, $P = .003$), worse functional status (91.6% vs. 48.5%, $P < .001$), and lower level of creatinine (1.4 mg/dl vs. 1.5 mg/dl, $P = .045$).

They were more likely to be ventilator dependent (52.6% vs. 2.9%, $P < .001$), treated with ECMO (36.5% vs. 2.0%, $P < .001$), IABP (27.7% vs. 7.5%, $P < .001$), and LVAD (7.9% vs. 2.2%, $P < .001$), require inotrope support (69.3% vs. 43.0%, $P < .001$), and ICU support (90.5% vs. 35.8%, $P < .001$). The donors for patients receiving early cardiac re-transplantation were less likely to have diabetes than donors for the late cardiac re-transplantation patients (0% vs. 3.2%, $P = .032$).

Among the early cardiac re-transplantation patients, the late pre-policy change group had lower tBil levels (1.1 mg/dl vs. 1.6 mg/dl, $P = .036$), were more often female (52.1% vs. 31.1%, $P = .020$), and more likely to have a history of malignancy (14.6% vs. 2.7%, $P = .015$) compared to the early pre-policy change group (Table S3). The recipients in the late pre-policy change group were more likely to be treated with ECMO (45.8% vs. 20.3%, $P = .003$) and disable or worse functional status (97.9% vs. 79.4%, $P = .006$), and less likely to have PRA > 10 (56.5% vs. 100%, $P = .003$) (Table S3).

For the early cardiac re-transplantation patients, the donors in the late pre-policy change group had a different distribution of cause of death ($P = .006$) and were more often CDC HR (27.1% vs. 3.0%, $P = .005$) compared to those for the early pre-policy change group (Table S4).

In the early cardiac re-transplantation patients, compared to the late pre-policy change group, the recipients in the post-policy change group were more likely to be treated with ECMO (86.7% vs. 45.8%, $P = .006$) (Table S3). The donors in the post-policy change group had a higher Donor risk index (6 vs. 4, $P = .032$) and were more often race mismatch (80% vs. 45.8%, $P = .021$) (Table S4).

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TABLE 1 Baseline characteristics of transplant recipients

	All cohort		Early retransplant		Late retransplant		
Total	***N* = 1294**		***N* = 137**		***N* = 1157**		***P* value**
Age at transplant (years)	45	[30–57]	50	[36–59]	44	[29–57]	.004[a]
Gender Female	460	35.5	52	38.0	408	35.2	.534
BMI at listing (kg/m^2)	25.8	[22.5–29.5]	25.4	[22.9–33.8]	25.8	[22.4–29.5]	.843
Ethnicity							.679
White	897	69.3	95	69.3	802	69.3	
Black	214	16.5	26	19.0	188	16.2	
Hispanic	126	9.7	9	6.6	117	10.1	
Asian	40	3.0	5	3.7	35	3.0	
Other	17	1.3	2	1.5	15	1.3	
Number of previous Tx (≥2)	50	3.9	5	3.6	45	3.9	.891
History of malignancy at listing	124	9.6	10	7.4	114	9.9	.342
Smoking history	237	21.9	27	26.7	210	21.5	.222
Diabetes at listing (Type 1 or Type 2)	316	24.6	29	21.2	287	24.8	.322
Creatinine at transplant (mg/dl)	1.5	[1.1–2.1]	1.4	[.9–2.1]	1.5	[1.1–2.1]	.045[a]
Creatinine at Transplant > 2 (mg/dl)	361	28.4	35	26.3	326	28.6	.585
Dialysis after listing	202	15.9	51	38.9	151	13.1	<.001[a]
tBil at transplant (mg/dl)	.8	[.5–1.2]	1.4	[.7–3.3]	.7	[.5–1.1]	<.001[a]
tBil at transplant > 2 (mg/dl)	151	12.0	48	36.6	103	9.1	<.001[a]
Need of transfusion after listing	294	23.8	90	69.8	204	17.6	<.001[a]
ECMO at transplant	73	5.6	50	36.5	23	2.0	<.001[a]
IABP at transplant	125	9.7	38	27.7	87	7.5	<.001[a]
LVAD at transplant	34	2.8	10	7.9	24	2.2	.002[a]
TAH at transplant	31	2.6	2	1.6	29	2.7	.444
Ventilator dependent at transplant	105	8.1	72	52.6	33	2.9	<.001[a]
Inotrope dependent at transplant	593	45.8	95	69.3	498	43.0	<.001[a]
PCWP at transplant (mm Hg)	17	[12–22]	19	[14–24]	17	[12–22]	.083
PVR at transplant (Wood units)	1.5	[1.0–2.3]	2.1	[1.2–2.9]	1.5	[1.0–2.2]	.003[a]
PRA > 10	428	76.3	24	60	414	77.5	.012[a]
Disabled or worse functional status	589	52.2	87	91.6	502	48.5	<.001[a]
ICU stay at transplant	538	41.6	124	90.5	414	35.8	<.001[a]
Transplant year							.002[a]
Early pre-policy change	532	41.1	74	54.0	458	39.6	
Late pre-policy change	634	49.0	48	35.0	586	50.7	
Post-policy change	128	9.9	15	11.0	113	9.8	

Continuous variables presented as median and interquartile range (IQR). Categorical variables presented as *n* (%).

[a]Statistically significant ($P < .05$).

Abbreviations: BMI, body mass index; ECMO, extracorporeal membrane oxygenation; IABP, intra-aortic balloon pump; ICU, intensive care unit; PCWP, pulmonary capillary wedge pressure; PRA, panel reactive antibody; PVR, pulmonary vascular resistance; LVAD, left ventricular assist device; TAH, total artificial heart; tBil, total bilirubin; Tx, transplantation.

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TABLE 2 Baseline characteristics of transplant donors

	All cohort		Early retransplant		Late retransplant		
Total	***N* = 1294**		***N* = 137**		***N* = 1157**		***P* value**
Donor age (years)	30	[21–40]	30	[22–41]	29	[21–40]	.387
Donor age, ≥ 50 (years)	106	8.2	13	9.5	93	8.0	.558
Donor gender female	457	64.7	46	33.6	411	35.5	.652
BMI at Transplant (kg/m^2)	25.6	[22.4–29.4]	25.9	[23.5–30.8]	25.6	[22.3–29.3]	.079
Donor ethnicity							.297
White	789	61.0	90	65.7	699	60.4	
Black	177	13.7	20	14.6	157	13.6	
Hispanic	273	21.1	24	17.5	249	21.5	
Asian	28	2.1	0	0	28	2.4	
Other	27	2.1	3	2.2	24	2.1	
Cause of death							.623
Anoxia	314	24.3	29	21.2	285	24.6	
CVA/stroke	236	20.6	32	23.4	234	20.2	
Head trauma	674	52.0	72	52.6	602	52.0	
CNS tumor	10	.8	2	1.5	8	.7	
Other	30	2.3	2	1.5	28	2.4	
Donor risk index	4	[3–6]	4	[2–6]	4	[3–5]	.515
Donor risk index ≥ 7	251	20.1	29	23.2	222	19.8	.375
Donor BUN/Cre ≧ 30	149	11.5	17	12.5	132	11.4	.709
Race mismatch	649	50.2	64	46.7	585	50.6	.395
Size mismatch	505	39.0	49	35.8	456	39.4	.408
Hepatitis C-positive	23	1.8	2	1.5	21	1.8	.766
CDC high risk donor	188	17.2	22	22.9	166	16.7	.125
Cocaine or other intravenous drug use	580	45.2	55	40.7	525	45.8	.267
Ejection fraction < 45%	11	.9	3	2.2	8	.7	.069
Donor diabetes mellitus	38	3.0	0	0	38	3.2	.032[a]
Donor hypertension	141	11.0	10	7.3	131	11.3	.147
Donor smoking history	192	15.0	22	16.4	170	14.7	.621
Ischemic time, (h)	3.3	[2.5–3.9]	3.3	[2.5–3.9]	3.3	[2.5–3.9]	.823
Ischemic time, ≧ 4.0 (h)	288	23.1	28	22.2	260	23.2	.802

Continuous variables presented as median and interquartile range (IQR). Categorical variables presented as *n* (%).
[a]Statistically significant ($P < .05$). Mismatch refers to a difference of 20% in weight between donor and recipient.
Abbreviations: BMI, body mass index; BUN, blood urea nitrogen; CDC, Centers for Disease Control; CNS, central nervous system; Cre, creatinine, CVA, cerebrovascular accident.

Among the late cardiac re-transplantation patients, the late pre-policy change group had lower tBil levels (.7 mg/dl vs. .8 mg/dl, $P < .001$) and were more often female (39.3% vs. 28.6%, $P < .001$) compared to the early pre-policy change group (Table S5). The recipients in the late pre-policy change group were more likely to be treated with IABP (7.5% vs. 3.1%, $P = .002$), LVAD (3.4% vs. .5%, $P = .004$), and TAH (3.9% vs. .8%, $P = .004$), and require ICU support (39.1% vs. 28.0%, $P < .001$). They were more likely to be disabled or have worse functional status (59.6% vs. 29.4%, $P < .001$), and less likely to have PRA > 10 (77.4% vs. 97.1%, $P < .001$) (Table S5).

For the late cardiac re-transplantation group, the donors in the late pre-policy change group had a different distribution of cause of death ($P < .001$), and more often had donor BUN/Cre ≧ 30 (12.8% vs.7.0%, $P = .002$), CDC HR (19.8% vs. 5.1%, $P < .001$), and cocaine or other intravenous drug use (52.7% vs. 33.9%, $P < .001$), and less likely to have smoking history (10.3% vs. 22.4%, P,.001) compared to the early pre-policy change group (Table S6).

In the late cardiac re-transplantation group, compared to the late pre-policy change group, the post-policy change group were less likely to have transfusion after listing (10.8% vs. 19.8%, $P = .025$), and

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TABLE 3 Incidence of post-transplant outcomes for early and late retransplant

	Early retransplant		Late retransplant		
Total	***N* = 137**		***N* = 1157**		***P* value**
Stroke	8	6.3	34	3.0	.052
Need of dialysis	56	42.4	222	19.6	<.001[a]
Need of permanent pacemaker	0	0	31	2.7	.059
Length of stay (days)	21	[11–47]	15	[10–24.3]	.001[a]
Acute rejection episode					
Treated with anti-rejection agent at least once	8	7.8	88	8.8	.721
30-day mortality	34	24.8	67	5.8	<.001[a]
1-year mortality	55	40.6	152	13.6	<.001[a]

Continuous variables presented as median and interquartile range (IQR). Categorical variables presented as *n* (%).
[a]Statistically significant ($P < .05$).

inotrope support (31.9% vs. 46.1%, $P = .005$), and more likely to be treated with ECMO (5.3% vs. 2.1%, $P = .045$), and IABP (25.7% vs. 7.5%, $P < .001$), and require ICU support (50.4% vs. 39.1%, $P = .025$) (Table S5). The recipients in the post policy-change were less likely to have PRA > 10 (55.1% vs. 77.4%, $P < .001$) (Table S5).

For the patients receiving late cardiac re-transplantation, the donors in the post-policy change group were more often Donor BUN/Cre ≧ 30 (22.1% vs. 12.8%, $P = .010$), Hepatitis C-positive (5.3% vs. 1.9%, $P = .030$), and CDC HR (31.0% vs. 19.8%, $P = .008$), and less likely to have diabetes mellitus (0% vs. 4.0%, $P = .033$) compared to those in the late pre-policy change group (Table S6).

4.2 | Early and long-term outcomes

When a re-transplant was performed within 1–1.5 years of the previous heart transplant, the probability of death at 1 year was higher than when a heart transplant was performed later than that. (Figure S2).

Table 3 compares re-transplant mortality and morbidity outcomes between the early and late re-transplantation groups. The early group had higher 30-days mortality (24.8% vs. 5.8%, $P < .001$), higher need of postoperative dialysis (42.4% vs. 19.6%, $P < .001$), and longer hospital stay (21 days vs. 15 days, $P = .001$).

There were 55 deaths within the first year in the early re-transplantation group and 152 deaths within the first year in the late retransplantation group (Figure 2). One-year mortality was significantly higher in the early transplantation compared with the late transplantation (Early 40.6% vs. Late 13.6%, log-rank $P < .001$). The follow up period for each was 383 days (IQR: 26–2183) in early retransplantation group and 1482 days (IQR: 373–2991) in the late retransplantation group.

In the early cardiac re-transplantation group, there was no significant difference in 1-year survival in each period (Early Pre-Policy Change 59.4%, Late Pre-Policy Change 63.9%, Post-policy Change 42.9%) (Figure S3).

In the late cardiac re-transplantation group, compared to the early pre-policy change group, 1-year survival was significantly higher in the late pre-policy change group (Early Pre-Policy Change 84.6% vs. Late Pre-Policy Change 87.8%, log-rank $P = .047$). There was no difference in 1-year survival between the late pre-policy change group and the post-policy change group (Late Pre-Policy Change 87.8% vs. Post-policy Change 87.7%, log-rank $P = .806$) (Figure S3).

Figure S4 shows that the ventilator dependence at transplant has a significant impact on 1-year mortality in both the early (76.2% vs. 44.4%; log-rank $P < .001$) and the late re-transplantation groups (87.4% vs. 53.3%; log-rank $P < .001$).

When patients with a baseline bilirubin > 2.0 are excluded, the early group had higher 30-day mortality (10.8% vs. 5.4%, $P < .001$), 1-year mortality (28.4% vs. 12.5%, log-rank $P < .001$), need for postoperative dialysis (31.3% vs. 18.7%, $P = .007$), and longer hospital stay (22 days vs. 15 days, $P < .001$) (Table S7). Recipients receiving the early cardiac re-transplantation with bilirubin > 2.0 had the worst 1-year survival rate (41.7%), compared to those in the early re-transplantation group with a tbil ≤ 2 or the late re-transplantation group (Figure S5).

4.3 | Random survival forests-ranked variable importance (RSF-ranked VIMP)

The results of RSF are shown in Figures 3 and 4. Total bilirubin (> 2 mg/dl) (VIMP, 2.99%) was an independent predictor of 1-year mortality after early re-transplantation and BMI (> 30.0 kg/m^2) (VIMP, 1.43%) and ventilator dependence (VIMP, 1.47%) were independent predictors for after late re-transplantation.

4.4 | Classical Cox modeling

Multivariable Cox modeling of recipients in the early re-transplantation group showed that recipient age, dialysis after listing,

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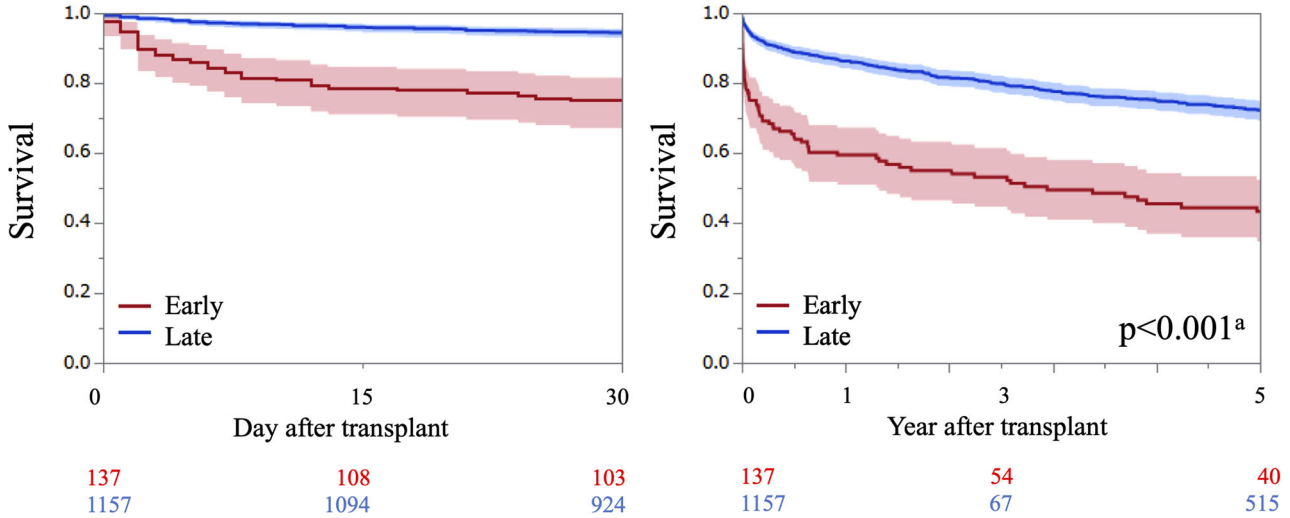

FIGURE 2 Kaplan-Meier survival curves for all-cause mortality. Early re-transplantation includes patients retransplanted within 1 year. Late re-transplantation includes patients includes patients retransplanted 1 year after previous transplant

and tBil (≧ 2.0 mg/dl) were associated with an increased risk of 1-year mortality (Table S8).

Univariable Cox models of 1-year mortality for recipients in the early re-transplantation, excluding both patients with total bilirubin ≥ 2.0 and ventilator dependence at the time of re-transplantation, showed that recipient age, recipient smoking history, dialysis after listing, donor risk index, race mismatch, female donor for male recipient, and donor smoking history were associated with an increased risk of 1-year mortality (Table S9).

Multivariable Cox modeling of recipients in the late re-trans plantation group showed that BMI (≧30.0 kg/m), tBil (≧ 2.0 mg/dl), blood transfusion after listing, ventilator dependent at transplant, and donor age were associated with an increased risk of 1-year mortality (Table S10).

5 | DISCUSSION

In our analysis, RSF showed that liver function was important in the early group, and obesity and preoperative mechanical ventilation were important in the late group. In addition to these factors, recipient age and dialysis in the early transplantation group, and liver dysfunction, need for transfusion, and donor age in the late transplantation group were identified as risk factors associated with 1-year mortality in the conventional Cox analysis.

The one novel observation of this study, in addition to those of two recent analyses of re-heart transplantation using UNOS,[6,7] is that a recipient total bilirubin levels greater than 2.0, indicating end-organ dysfunction, is an independent risk factor for 1-year mortality among those individuals receiving retransplantation within the first year after their original transplant. Patients who received retransplantation within 1 year of their previous transplant with bilirubin levels above 2.0 at transplant had a poorer 1-year survival rate (41%).

In the late cardiac re-transplantation patients, our study demonstrated a 3-fold increase in IABP use at the time of transplantation after the policy change. In contrast, LVAD use in patients with ACHD at the time of transplantation, which had been gradually increasing, substantially decreased after the policy change. Further observation is needed for the impact of this change on their outcomes.

Cardiac re-transplantation is generally associated with lower survival rates compared to primary transplantation. All patients undergoing cardiac re-transplantation have had a previous sternotomy, which increases the risk of mortality following transplant.[7,28,29] The interval from the previous transplant is a particularly important factor. Previous reports have shown that the outcomes of cardiac re-transplantation occurring greater than 1 year after primary transplantation are comparable to that of primary transplantation.[7,16] Our analysis revealed that, for the recent two decades, the patients receiving late cardiac re-transplantation had improved survival while those receiving early cardiac re-transplantation still have poor survival. The outcome of a short interval re-transplantation in our analysis was extremely poor, and it would be reasonable to consider it a contraindication, consistent with previous reports.[16]

Obesity, hepatic dysfunction, and the need for mechanical ventilation are well-known risk factors for cardiac transplantation. Overall, weight loss is recommended because pre-transplant obesity (BMI > 30 kg/m^2) is associated with poor outcome after heart transplantation.[24] Heart transplantation in obese patients is challenging because it makes the procedure and postoperative management more difficult and delays wound recovery.[30] The need for mechanical ventilation is a measure for poor cardiopulmonary reserve in patients.[31] Patients on ventilators tend to stay longer in hospitals and intensive care units, which can lead to poor outcomes owing to nosocomial infections, malnutrition, and generalized deterioration.[31] Hepatic dysfunction reflects severity of illness with end-organ dysfunction at the time of transplantation.[10] In our analysis, mechanical circulatory

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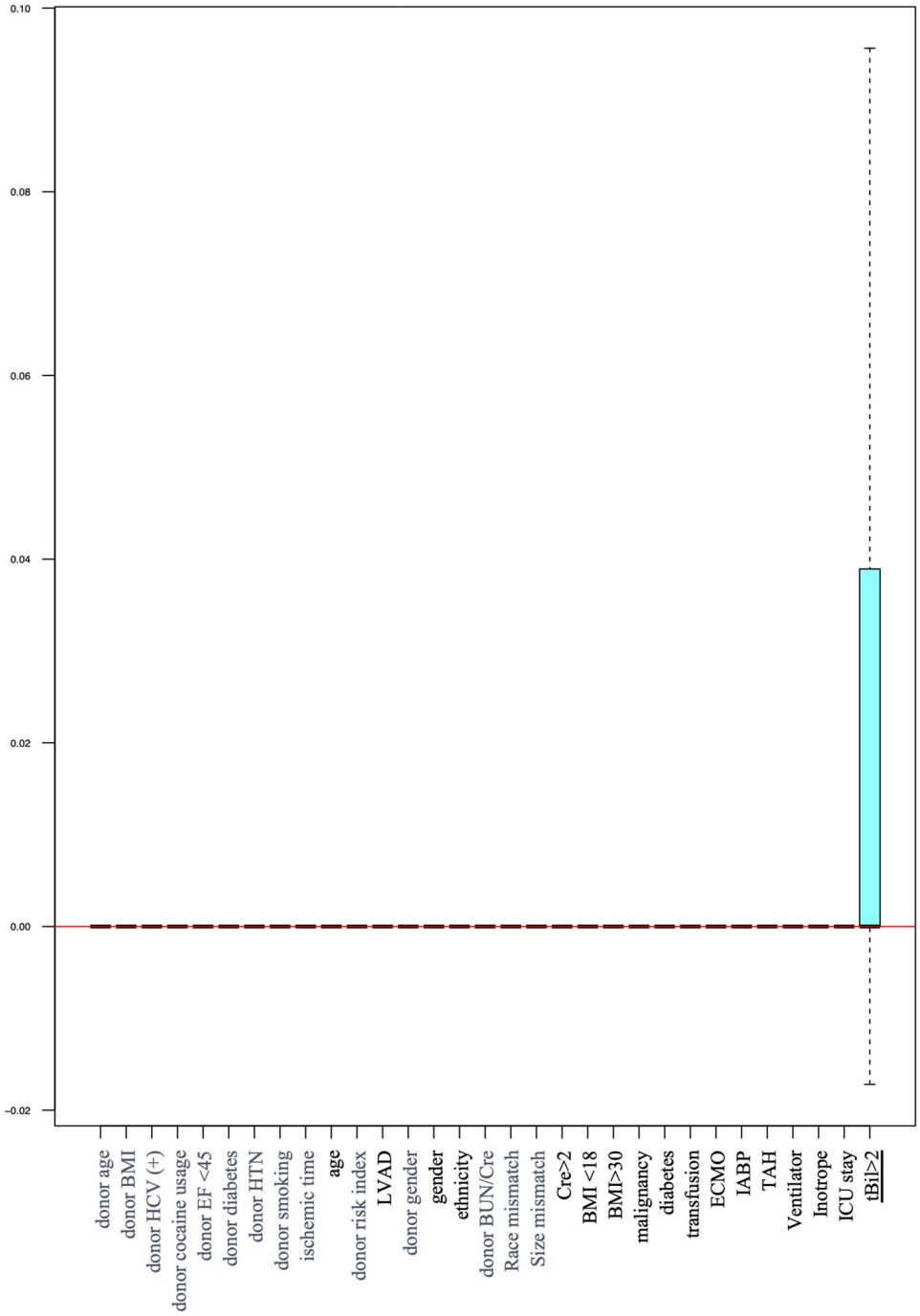

FIGURE 3 Box plot for variable importance predicting of 1-year mortality in the early cardiac re-transplantation. BMI, body mass index; BUN, blood urea nitrogen; Cre, creatinine; ECMO, extracorporeal membrane oxygenation; EF, ejection fraction; HTN, hypertension; IABP, intra-aortic balloon pump; ICU, intensive care unit; LVAD, left ventricular assist device; TAH, total artificial heart; tBil, total bilirubin; Tx, transplantation

supports such as ECMO, IABP, and LVAD were not risk factors in either group. Early referral and intervention to maintain stable peripheral circulation may lead to better results.

Our study indicates that recipient factors have a greater impact on survival than donor factors. As previous reports,[31,32] our analysis, which included donor factors, suggested that older donor age was associated with worse survival in re-transplantation. In studies of primary transplantation, the use of a heart from a donor aged 40 years or above was associated with increased mortality, independent of other risk factors.[33,34] In recent years, the upper age limit for cardiac donation was 50–55 years old, and above that age was considered a marginal donor.[20–23,35] However, if it is structurally normal, there is no conclusive reason why the selected older donor hearts cannot be used. Older donors with shorter ischemic times may have favorable clinical outcomes.[34] The potential benefit on survival of patients who received hearts from older donors must be compared with those who died while

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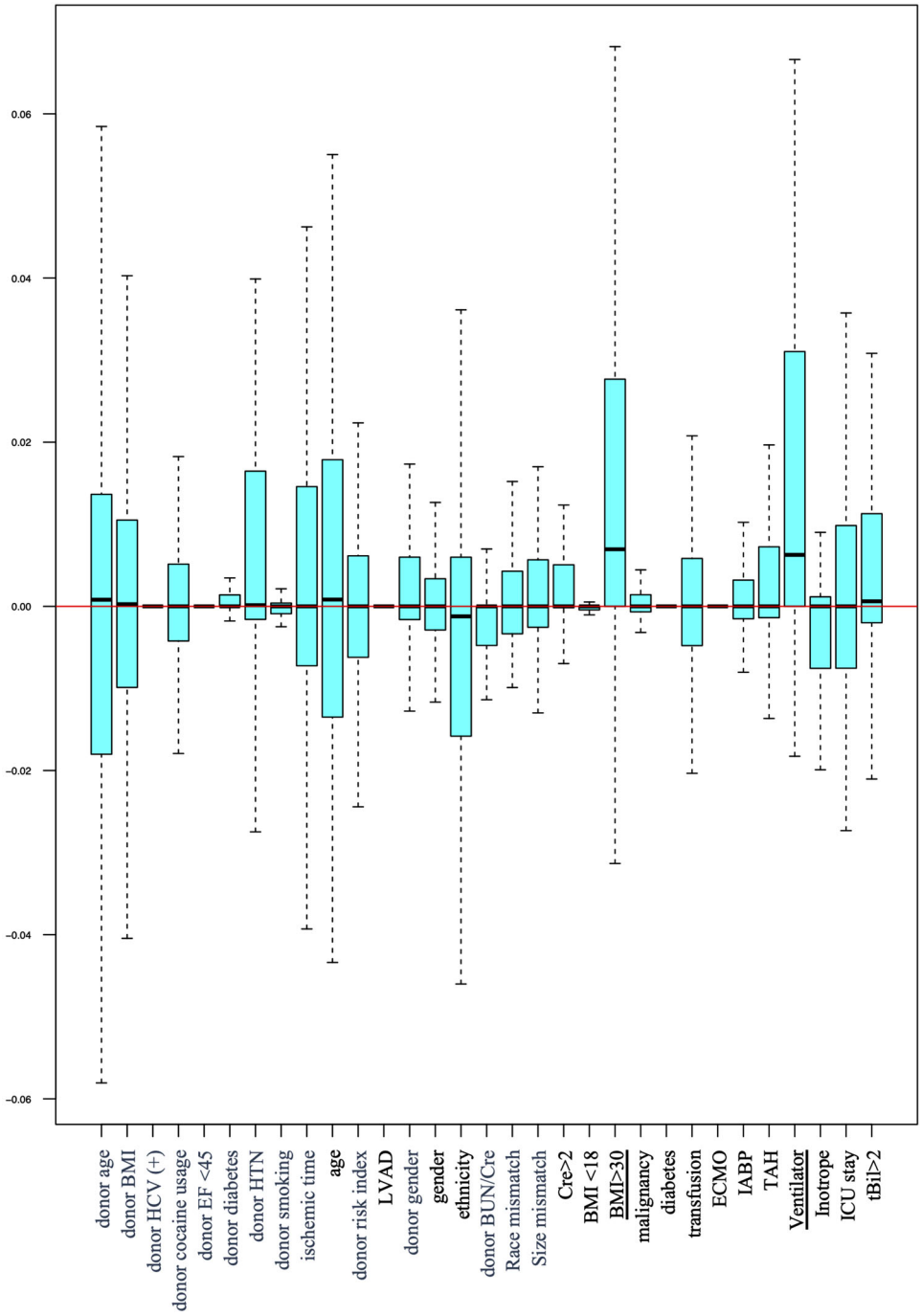

FIGURE 4 Box plot for variable importance predicting of 1-year mortality in the late cardiac re-transplantation. BMI, body mass index; BUN, blood urea nitrogen; Cre, creatinine; ECMO, extracorporeal membrane oxygenation; EF, ejection fraction; HTN, hypertension; IABP, intra-aortic balloon pump; ICU, intensive care unit; LVAD, left ventricular assist device; TAH, total artificial heart; tBil, total bilirubin; Tx, transplantation

waiting without receiving organ. Our data suggest that recipient factors may have a greater influence on survival rather than donor factors in the re-transplant group. In this study, the use of marginal donors such as hepatitis C-positive, cocaine or intravenous drug use, and CDC HR had increased in recent years. Our data support the concept that heart re-transplantation in older donors should be considered under certain conditions, given the shortage of organs, and the notion that cardiac re-transplantation using a marginal donor should be considered, albeit cautiously, in view of organ shortages and the ethics of re-transplantation. It is preferable in terms of donor pooling to utilize such donors effectively, considering the facility and the candidate's conditions.

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6 | LIMITATIONS

There are several limitations in this study. This study was retrospective, so some preoperative information in UNOS data is limited and sometimes missing. As in previous studies,[7] there were high rates of missing data. For example, the PRA showed 50% missing. Even though, random forests based on multiple input data are less affected by the occurrence of missing values,[36] factors with high missing rates (> 10%) were excluded from the current analysis, which may have affected the results. Our UNOS data set does not consider the primary diagnosis of heart transplantation. Indication for retransplantation,[8,16] and the volume of the transplant center,[37,38] may have also affected the prognosis.

7 | CONCLUSION

We evaluated 1294 cardiac re-transplantation patients from the UNOS registry to identify factors that significantly increased risk of 1-year mortality. Cardiac re-transplantation within 1 year after primary transplantation was associated with higher 1-year mortality than cardiac re-transplantation after 1 year. Liver dysfunction, obesity, and ventilator dependence at transplant were considered risk factors by machine learning. Cox analysis also identified recipient and donor age, renal function, and need for transfusion as additional risk factors. Adequate patient selection and proper preoperative management are necessary in these challenging patients. If certain criteria are met, cardiac re-transplantation may lead to a better prognosis.

CONFLICT OF INTEREST

Y. Naka has received consulting fee from Abbott Incorporated. The remaining authors have no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Atsushi Kainuma https://orcid.org/0000-0003-0986-3446
Amy S. Wang https://orcid.org/0000-0002-5970-0517

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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